

Aditya Sneh

AI Engineer

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Experience

Project Junior Research Fellow

Aug 2024 – Jul 2025

Systems and Informatics Research Laboratory

Bhopal, India

- Built **LogMe**, a 24/7 Android data-collection app capturing **23,520+ hours** of IMU and smartphone usage data from 70 users using FastAPI and PostgreSQL.
- Engineered a Python pipeline for 13 IMU streams, applying filtering, 50-to-40 Hz downsampling, and overlapping 2.5s segmentation to generate **DySTAN** training windows.
- Architected DySTAN, a PyTorch CNN-BiLSTM model for sedentary activity and social-context classification, achieving **0.83 joint accuracy**, **18.7% above** the strongest baseline.
- Implemented on-device **Thompson Sampling** across 10 activity contexts and 16 interventions, outperforming the 0.5 random baseline in real-world deployment.
- Deployed a clinician-informed JITAI with an AIIMS psychiatrist, logging **10,687 prompts** at a **76.4% response rate** for context-aware mental-health interventions.

AI/ML Subject Matter Expert

Apr 2024 – Jul 2024

Grades Buddy, Codepedia Solutions Pvt. Ltd.

Remote

- Built scikit-learn pipelines in Python across **12K+ samples**, improving macro-F1 by **14%** over baseline.
- Trained PyTorch deepfake detectors, applied INT8 quantization, and deployed optimized inference while retaining a **0.88 F1 score**.
- Fine-tuned **Llama 2** and **Mistral 7B** with LoRA/PEFT and built embedding-based RAG pipelines, achieving **85% task accuracy** and **0.82 faithfulness**.
- Containerized and deployed ML models on AWS with Docker, tracking **50+ experiments** through MLflow and automated CI/CD.

Technical Skills

Languages: Python, SQL, TypeScript

Data & Databases: Pandas, NumPy, PostgreSQL, FAISS, Pinecone, Chroma

Machine Learning & Deep Learning: scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers

LLM Systems: Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-Tuning (LoRA, PEFT), LangChain, LangGraph, OpenAI API, Agentic AI, Multi-Agent Systems, Model Evaluation

Cloud, MLOps & Developer Tools: AWS, GCP, Docker, MLflow, GitHub Actions, Git, CI/CD, Monitoring & Observability

Application Development: FastAPI, React, Next.js

Education

B.S. & M.S., Data Science and Engineering

2019 – 2024

Indian Institute of Science Education and Research (IISER)

Bhopal, India

CGPA: 8.26/10

Publications

- HCFSLN: Adaptive Hyperbolic Few-Shot Learning for Multimodal Anxiety Detection. *ICMI*, 2026.
- On the Robustness of Iris Presentation Attack Detectors. *BMVC*, 2025.
- DySTAN: Joint Modeling of Sedentary Activity and Social Context from Smartphone Sensors. *arXiv*, 2025.
- Fairness-Aware Few-Shot Learning for Audio-Visual Stress Detection. *arXiv*, 2025.
- Real-World Receptivity to Adaptive Mental Health Interventions. *arXiv*, 2025.

Projects

BloomPilot: Multi-Agent AI Plant Care Platform | [Live Demo](#) Jun 2026 – Jul 2026

LangGraph, Multi-Agent Orchestration, Next.js, TypeScript, OpenAI

- Architected a **7-node LangGraph pipeline** with a ReAct planner orchestrating **5 tools** over **10 iterations (~13s)**, grounding action plans in live weather, species, and care data.
- Built an automated evaluation harness across **6 locations**, verifying complete **7-agent traces**, **15 grounded references** per plan, and deterministic outputs.
- Engineered agent guardrails for duplicate calls, timeouts, and confidence-gated delivery (**0.65+**), independently validating outputs against live weather and timezone data.
- Shipped a multi-channel (push, email, Telegram) **notification engine** with suppression, escalation, and deduplication logic, cutting redundant notifications to zero in production testing

MemoStack: Multimodal RAG for Enterprise Decisions | [Live Demo](#) Jul 2025 – Sep 2025

Pinecone, PostgreSQL, CI/CD, Render, Vercel

- Built a multi-tenant RAG system across 6 enterprise sources, delivering tenant-isolated, evidence-cited answers.
- Designed hybrid retrieval with dense search, BM25 fusion, and cross-encoder reranking, achieving **0.80 Recall@1**, **0.85 Recall@5**, and **0.956 MRR** on 210 QA cases across 50 documents.
- Implemented evidence-based abstention, correctly rejecting **10/10 unsupported queries** to avoid ungrounded answers.
- Added 4-layer caching, 2-model LLM routing, and tiered rate limits; benchmarked the retrieval path at **1.06K queries/sec with p95 latency below 184 ms** across 16 workers.

MindPulse: On-Device Behavioral Drift Detection & Intervention | [Live Demo](#) Mar 2026 – Jul 2026

PyTorch, LLM Fine-Tuning, LoRA, Quantization

- Built a behavioral-sensing pipeline on smartphone data from **49 participants**, training a PyTorch temporal convolutional network with leave-one-participant-out cross-validation.
- Fine-tuned Gemma 3 270M with LoRA on **4,589 instructions**, recording zero policy violations across 565 held-out safety and format tests.
- Deployed Gemma 3 on-device via LiteRT-LM in an Android app with deterministic safety controls and personalized drift detection, achieving **2.6e-7 output deviation** and **sub-10-second inference**.